Supernetting

- An organization can apply for a set of class C blocks instead of just one.
- For example, an organization that needs 1000 addresses can be granted four contiguous class C blocks.
- The organization can then use these addresses to create one supernetwork.
- Supernetting decreases the number of Is in the mask.
- For example, if an organization is given four class C addresses, the mask changes from /24 to /22. We will see that classless addressing eliminated the need for supernetting

Super netting

Supernetting

a) Although class A and B addresses are almost depleted, Class C addresses are still available.

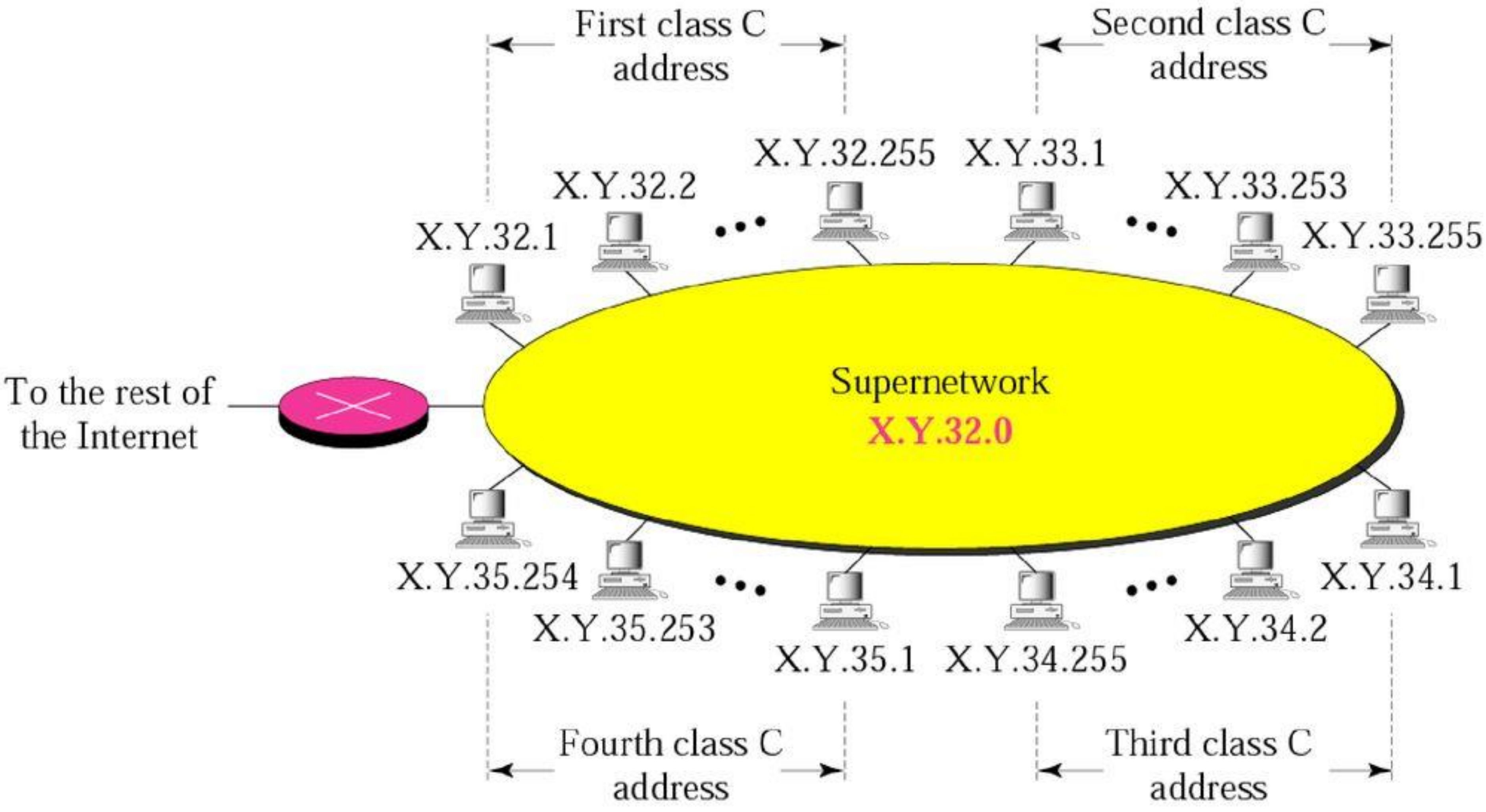
b) However, the size of a class C block with a maximum number of 256 addresses may not satisfy the needs of an organisation.

c) One solution is supernetting.

d) In supernetting, an organisation can combine several class C blocks to create a large range of addresses.

e) In order words, several networks are combined to create a supernetwork. This is done by applying a set of class C blocks instead of just one.

Example of a Supernetwork



Rules:

- The number of blocks must be a power of 2 i.e: (2, 4, 8, 16, . . .).
- The blocks must be contiguous in the address space (no gaps between the blocks).
- The third (3^{rd}) byte of the first (1^{st}) address in the superblock must be evenly divisible by the number of blocks. In other words, if the number of blocks is N, the third byte must be divisible by N.

Example

A company needs 600 addresses. Which of the following set of class C blocks can be used to form a supernet for this company?

a.	198.47.32.0	198.47.33.0	198.47.34.0	
b.	198.47.32.0	198.47.42.0	198.47.52.0	198.47.62.0
c.	198.47.31.0	198.47.32.0	198.47.33.0	198.47.52.0
d.	198.47.32.0	198.47.33.0	198.47.34.0	198.47.35.0

Solution

a: **No, there are only three blocks. (not to the power of 2)**

b: **No, the blocks are not contiguous.**

c: **No, 31 in the first block is not divisible by 4.**

d: **Yes, all three requirements are fulfilled.**

Vital notes: Supernetting

In subnetting,
we need the first address of the subnet
and the subnet mask to
define the range of addresses.

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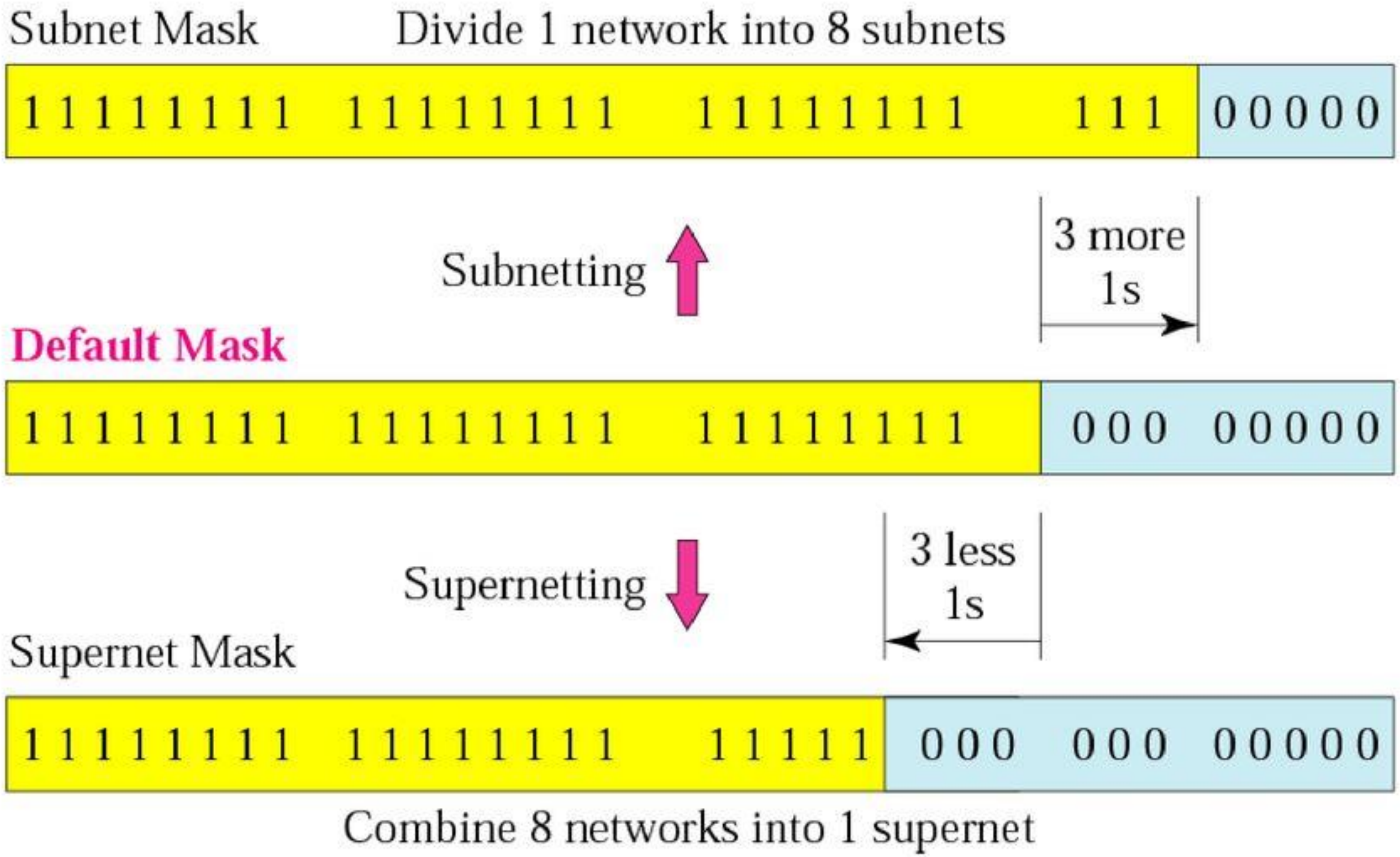
Comparison of subnet, default, and supernet masks



Example

We need to make a supernetwork out of 16 class C blocks. What is the supernet mask?

Solution

Class C mask is defaulted with 24 of ‘1’ is

11111111 11111111 11111111 00000000

We need 16 blocks. For 16 blocks we need to change four 1s to 0s in the default mask. So the mask is

11111111 11111111 1111**0000** 00000000

or

255.255.240.0

Example

A supernet has a first address of 205.16.32.0 and a supernet mask of 255.255.248.0. A router receives 3 packets with the following destination addresses:

205.16.37.44

205.16.42.56

205.17.33.76

Q: Which packet belongs to the supernet?

Solution

We apply the supernet mask to find the beginning address.

205.16.37.44 AND 255.255.248.0 → 205.16.32.0

205.16.42.56 AND 255.255.248.0 → 205.16.40.0

205.17.33.76 AND 255.255.248.0 → 205.17.32.0

Only the first address belongs to this supernet.

Example

A supernet has a first address of 205.16.32.0 and a supernet mask of 255.255.248.0. How many blocks are in this supernet and what is the range of addresses?

Solution

The supernet mask has 21 ‘1’s. The default mask has 24 1s. Since the difference is 3, there are 2^3 or 8 blocks in this supernet. The blocks are 205.16.32.0 to 205.16.39.0. The first address is 205.16.32.0. The last address is 205.16.39.255.